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Cicla EcoPod – Power Consumption and Autonomous Solar System Report

M-CCIA Project I – Milestone 1 Technical Annex

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1. System Overview

EcoPod is an autonomous, solar-powered, smart bicycle parking pod for 12 bicycles located in Lisbon, Portugal. The system operates off-grid using solar panels, LiFePO4 batteries and MPPT charge controller. All components are selected for low power consumption and high reliability.

2. Power Consumption Calculation

Voltage reference: 12 V nominal (DC-DC conversion efficiency 90 %).

Location solar insolation (Lisbon): average 4.1 kWh/m²/day (winter 2.7, summer 7.7).

Component per bike stand (×12)	Qty	Stand-by (mA)	Active (mA)	Duty cycle	Daily energy (mAh/day)
ESP32 (deep-sleep + periodic wake)	12	0.01	200	1 min/h	4 080
LED strip 1 m white + 1 m RGB (CRI>90, 120 LED/m)	12	–	960	2 h/night	23 040
2 × solenoid (normally closed, 12 V)	24	0	500	10 s × 20/day	5 556
2 × servo (height + clamp)	24	100	500	10 s × 20/day	2 400 + 5 556
Hall sensor + limit switches + button	–	<6	–	100 %	1 728
Pressure/tension sensors (2×)	24	0.5	–	100 %	288
RFID module (MFRC522)	12	20	–	100 %	5 760
Subtotal per 12 stands					48 412 mAh/day

Pod-wide components	Qty	Stand-by (mA)	Active (mA)	Duty cycle	Daily energy (mAh/day)
Wi-Fi access point / ESP-NOW mesh	1	200	–	100 %	4 800

Pod-wide components	Qty	Stand-by (mA)	Active (mA)	Duty cycle	Daily energy (mAh/day)
Weather station (DHT22 + anemometer + rain)	–	35	–	100 %	840
Ventilation fans (covered, low-speed)	1	–	500	4 h/day	2 000
OLED display 0.96" + intercom standby	1	70	–	100 %	1 680
Motorised door lock + motion sensors	–	<10	200	5 s × 20/day	240 + 67
Emergency lights + intercom battery set	–	0	20	backup only	<100
Subtotal pod-wide					9 627 mAh/day

Total daily consumption (12 V):

- Stand-by / low-duty: ≈ 1.9 Ah
- High-duty (LED + fans): ≈ 2.1 Ah
- **Grand total:** 4.0 Ah/day (48 Wh/day) conservative estimate
- Measured peak current during simultaneous operation: < 3 A

3. Solar Power System Specification (Off-Grid, Lisbon)

Item	Specification	Daily yield (Lisbon)	Notes
Solar panel	120 W monocrystalline 12 V (0.8 m ²)	490–550 Wh/day	Roof-integrated, tilt 30°
Charge controller	MPPT 20 A (e.g., EPEVER Tracer 2210AN)	98 % efficiency	Low-voltage disconnect, temperature comp.
Battery bank	LiFePO ₄ 12.8 V 30 Ah (384 Wh usable)	3–5 days autonomy	3000+ cycles, built-in BMS
Supercapacitor buffer (per ESP32)	5 V 1–5 F	–	Eliminates brown-out during Wi-Fi TX

System efficiency: 82 % (panel → battery → load).

Winter worst-case autonomy: 3.2 days at 2.7 kWh/m²/day insolation.

Energy balance: 490 Wh produced vs 48 Wh consumed → 10× margin.

4. Implemented Optimisations

- ESP32 deep-sleep + ULP co-processor for sensor polling (consumption < 15 μ A).
- ESP-NOW mesh network replaces dedicated Wi-Fi router (–300 mA).
- LED PWM dimming 30–50 % brightness at night (–50 % energy).
- Solenoids and servos energised only on confirmed RFID + user action.
- Separate NiMH 12 V 2 Ah pack for emergency lights and intercom (isolated).
- Predictive AI model (Applied Machine Learning) disables ventilation and reduces LED duty when low occupancy is forecasted.

5. Additional Technical Enhancements

- Ultrasonic sensors (HC-SR04) added for precise bicycle detection and 3D back-office visualisation in Unity.
- Perimeter PIR + ultrasonic array for placement analytics and anti-tampering.
- Real-time battery SoC monitoring via ESP32 ADC → p5.js dashboard.
- CO₂ savings counter displayed on pod OLED and back-office (Creative Programming).
- Central Raspberry Pi gateway option for multiple pods (future scaling).

The proposed power architecture guarantees full off-grid operation in Lisbon climate, meets all prototype requirements for Milestone 2 and 3, and provides sufficient headroom for additional sensors or future features.